

COSMOLOGY

I. Motivation: Cosmological Correlations

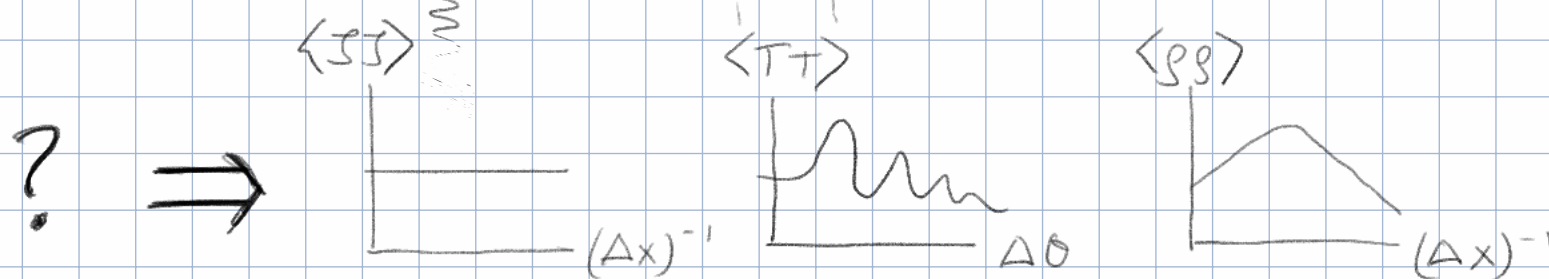
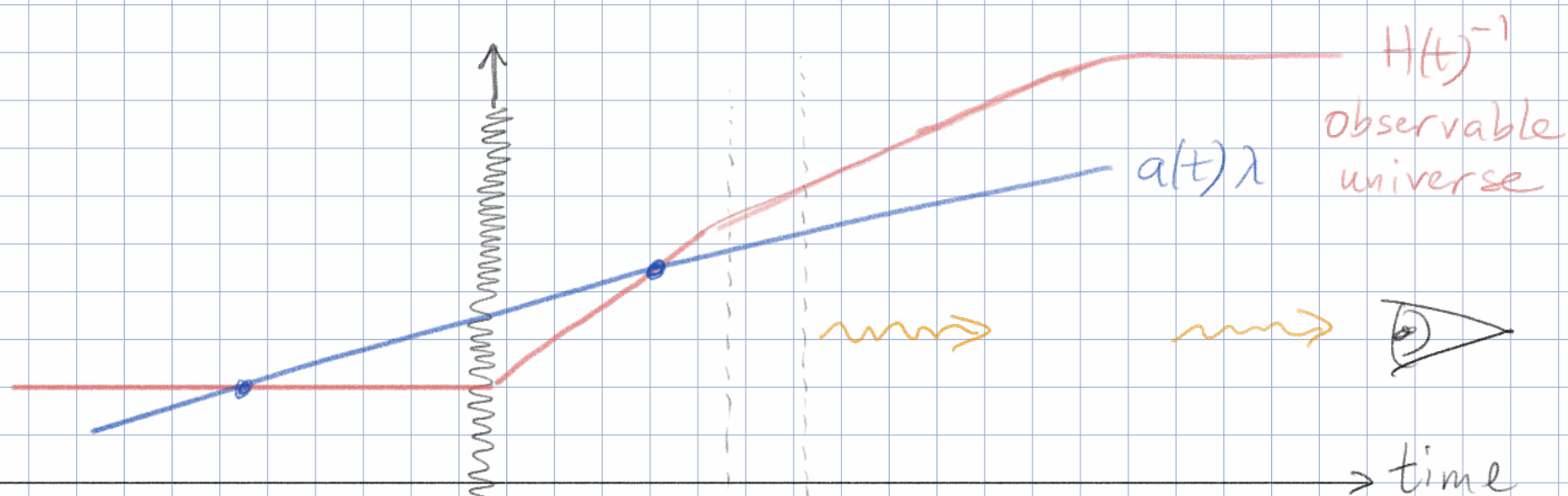
II. Inflation and Initial Conditions

III. Correlators

IV. Wave function

V. Cosmological Bootstrap

I. MOTIVATION: The Big Picture



Observed correlations can be traced back in time:

$$\delta \rho_A(t, \underline{x}) = \int \frac{d^3 k}{(2\pi)^3} e^{i \underline{k} \cdot \underline{x}} T_k^{(A)}(t, t_i) J_{\underline{k}}(t_i) \quad A = \rho, b, m, \dots$$

$$\left\langle \prod_{a=1}^N \delta \rho_A(t, \underline{k}_a) \right\rangle = \left[\prod_{b=1}^N T_{\underline{k}_b}^{(A)}(t, t_i) \right] \left\langle \prod_{c=1}^N J_{\underline{k}_c}(t_i) \right\rangle$$

OBSERVED CORRELATIONS

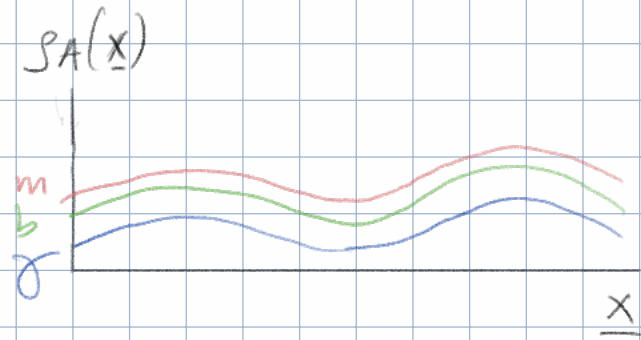
TRANSFER FUNCTION

INITIAL CONDITIONS

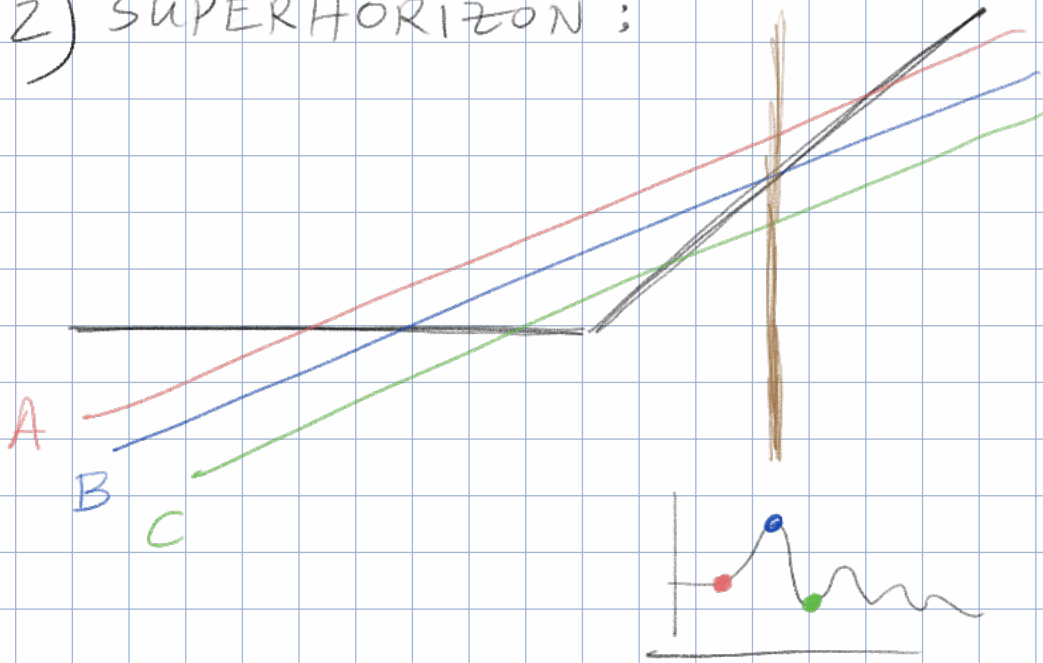
Four facts about initial conditions:

1) ADIABATIC:

$$\frac{\delta \rho_A}{\bar{\rho}_A + \rho_A} = \frac{\delta \rho_B}{\bar{\rho}_B + \rho_B} \equiv \mathcal{I}(\underline{x}) \quad \forall A, B$$



2) SUPERHORIZON:



3) SCALE INVARIANCE

POWER SPECTRUM

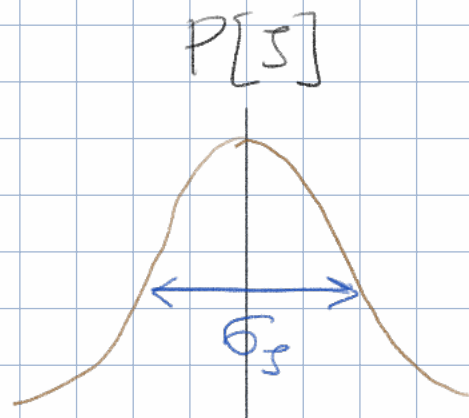
$$\langle \mathcal{I}_{\underline{k}} \mathcal{I}_{\underline{k}'} \rangle = (2\pi)^3 \delta(\underline{k} + \underline{k}') P(\underline{k})$$

$$P(\underline{k}) = \frac{A}{k^{3+0.03}} \Rightarrow \langle \mathcal{I}(\underline{x}) \mathcal{I}(\underline{x}') \rangle \propto \text{const} \equiv 6_s^2$$

$$\text{cf. } \langle \mathcal{I}(\underline{x}) \mathcal{I}(\underline{x}') \rangle_{\text{Mink}} = \frac{1}{|\underline{x} - \underline{x}'|^2}$$

4) GAUSSIAN

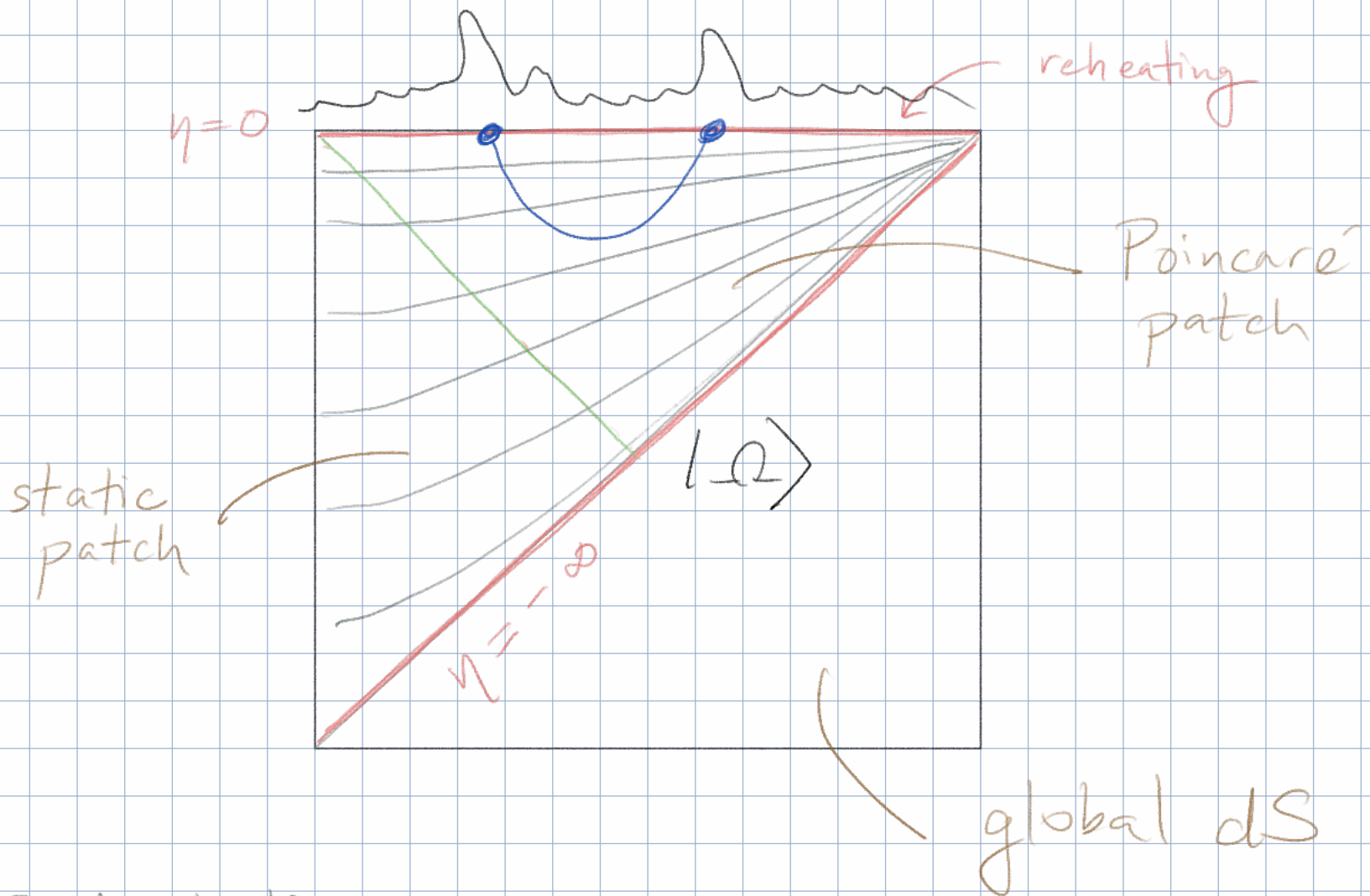
$$f_{NL} \sim \frac{\langle \mathcal{I} \mathcal{I} \mathcal{I} \rangle}{\langle \mathcal{I} \mathcal{I} \rangle^{3/2}} \lesssim 10$$



II. INFLATION AND INITIAL CONDITIONS

- FRW metric: $ds^2 = -dt^2 + \underbrace{a^2(t)}_{\text{SCALE FACTOR}} d\underline{x}^2$
- Inflation = accelerated expansion $\ddot{a} > 0$
 $\approx$ de Sitter spacetime $a(t) = e^{Ht}$

$$ds^2 = \frac{-d\eta^2 + d\underline{x}^2}{(H\eta)^2}, \quad \eta \equiv \int \frac{dt}{a(t)} \quad \text{conformal time}$$



- Perturbations:

$$\phi(\eta, \underline{x}) \equiv \bar{\phi}(\eta) + \delta\phi(\eta, \underline{x})$$

$$ds^2 = -N^2 dt^2 + N_i dt dx^i + a^2(t) e^{2\mathcal{J}(t, \underline{x})} \delta_{ij} dx^i dx^j$$

- Action (in comoving gauge):

$$S_2 = \frac{1}{2} \int d\eta d^3x \, 2a^2 \epsilon \left[(\mathcal{J}')^2 - (\nabla \mathcal{J})^2 \right]$$

$\epsilon \equiv -\dot{H}/H^2$

- Action (in spatially flat gauge)

$$S_2 = \frac{1}{2} \int d\eta d^3x \, a^2 \left[(\delta\phi')^2 - (\nabla \delta\phi)^2 + 3\epsilon H^2 (\delta\phi)^2 \right]$$

- QUANTIZATION:

Consider a spectator field $\phi(\eta, \underline{x})$ in exact dS

$$S = \frac{1}{2} \int d\eta d^3x \, a^2 \left[(\phi')^2 - (\nabla \phi)^2 - a^2 m^2 \phi^2 \right]$$

Define $v \equiv a\phi$, so that

$$\begin{aligned} S &= \frac{1}{2} \int d\eta d^3x \left[(v')^2 - (\nabla v)^2 + \left(\frac{a''}{a} - a^2 m^2 \right) v^2 \right] \\ &= \frac{1}{2} \int d\eta d^3x \left[(v')^2 - (\nabla v)^2 + \frac{2 - m^2/H^2}{\eta^2} v^2 \right] \end{aligned}$$

The equation of motion is

$$v_k'' + \left(k^2 + \frac{m^2/H^2 - 2}{\eta^2} \right) v_k = 0$$

Focus on $m=0$.

$$\text{Let } \hat{\phi}_{\underline{k}}(\eta) \equiv f_{\underline{k}}(\eta) \hat{a}_{\underline{k}} + f_{\underline{k}}^*(\eta) \hat{a}_{-\underline{k}}^+$$

$$\text{where } [\hat{a}_{\underline{k}}, \hat{a}_{\underline{k}'}^+] = (2\pi)^3 \delta(\underline{k} - \underline{k}') \text{ and } \hat{a}_{\underline{k}} |0\rangle = 0$$

VACUUM

$$f_{\underline{k}}(\eta) \equiv \frac{v_{\underline{k}}(\eta)}{a(\eta)}, \quad v_{\underline{k}}'' + \left(k^2 - \frac{2}{\eta^2}\right) v_{\underline{k}} = 0$$

Bunch-Davies mode function:

$$v_{\underline{k}}(\eta) = \frac{1}{\sqrt{2k}} \left(1 - \frac{i}{k\eta}\right) e^{-ik\eta} \xrightarrow{\eta \rightarrow -\infty} \frac{1}{\sqrt{2k}} e^{-ik\eta}$$

MINKOWSKI

$$f_{\underline{k}}(\eta) = \frac{iH}{\sqrt{2k^3}} (1 + ik\eta) e^{-ik\eta}$$

Zero-pt fluctuations:

$$\begin{aligned} \langle 0 | \hat{\phi}(\eta, \underline{x}) \hat{\phi}(\eta, \underline{x}') | 0 \rangle &= \int \frac{d^3 \underline{k}}{(2\pi)^3} e^{i\underline{k} \cdot \underline{x}} \int \frac{d^3 \underline{k}'}{(2\pi)^3} e^{i\underline{k}' \cdot \underline{x}'} \langle 0 | \hat{\phi}_{\underline{k}}(\eta) \hat{\phi}_{\underline{k}'}(\eta) | 0 \rangle \\ &= \int \frac{d^3 \underline{k}}{(2\pi)^3} \int \frac{d^3 \underline{k}'}{(2\pi)^3} e^{i\underline{k} \cdot \underline{x} + i\underline{k}' \cdot \underline{x}'} \langle 0 | (\hat{a}_{\underline{k}} f_{\underline{k}}(\eta) + \hat{a}_{-\underline{k}}^+ f_{\underline{k}}^*(\eta)) (\hat{a}_{\underline{k}'} f_{\underline{k}'}(\eta) + \hat{a}_{-\underline{k}'}^+ f_{\underline{k}'}^*(\eta)) | 0 \rangle \\ &= f_{\underline{k}}(\eta) f_{\underline{k}'}^*(\eta) \langle 0 | \underbrace{[\hat{a}_{\underline{k}}, \hat{a}_{-\underline{k}'}^+]}_{(2\pi)^3 \delta(\underline{k} + \underline{k}')} | 0 \rangle \\ &= \int \frac{d^3 \underline{k}}{(2\pi)^3} e^{i\underline{k} \cdot (\underline{x} - \underline{x}')} |f_{\underline{k}}(\eta)|^2 \end{aligned}$$

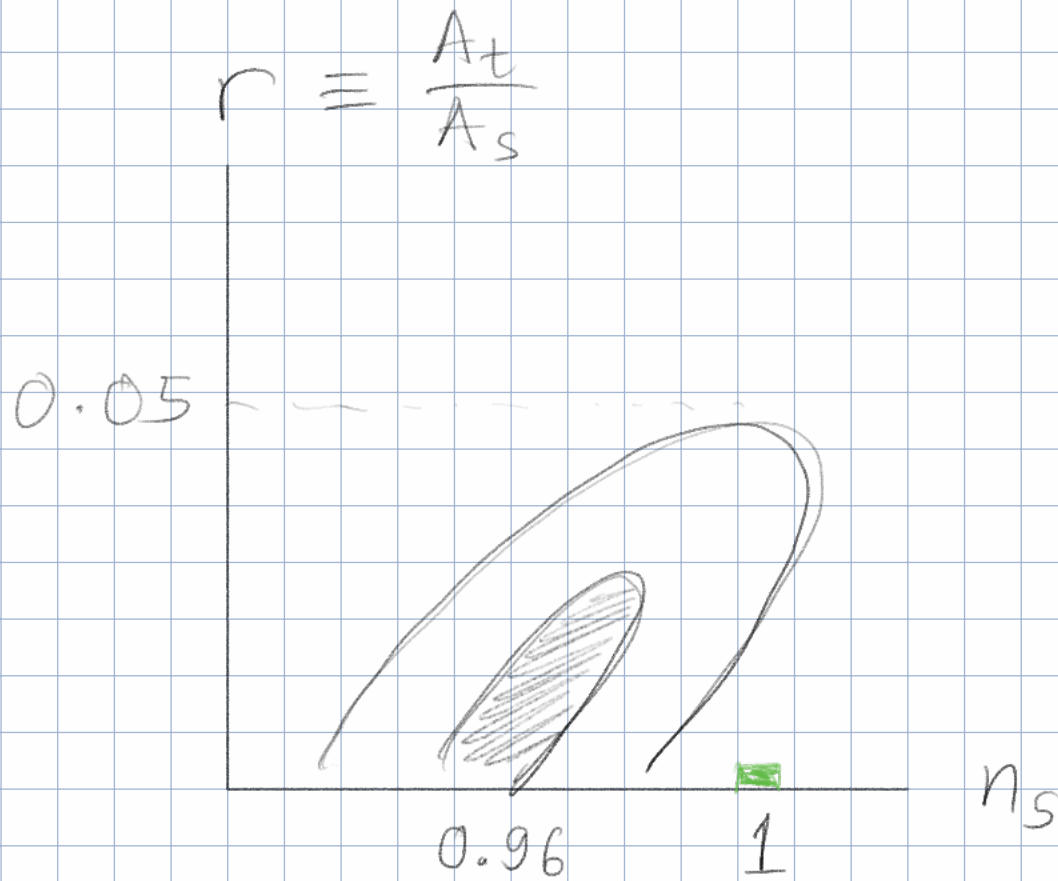
POWER SPECTRUM

$$P_{\phi}(k, \eta) \equiv |f_{\underline{k}}(\eta)|^2 = \frac{H^2}{2k^3} (1 + (k\eta)^2) \xrightarrow{\eta \rightarrow 0} \boxed{\frac{H^2}{2k^3}}$$

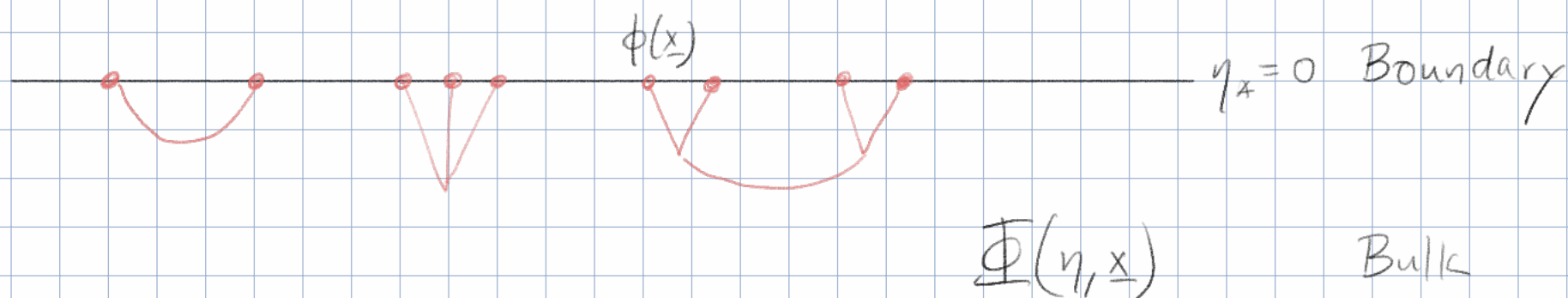
• INFLATIONARY POWER SPECTRA

Scalars : $\frac{k^3}{2\pi^2} P_g(k) = \frac{1}{8\pi^2 \epsilon} \frac{H^2}{M_{Pl}^2} \equiv A_s \left(\frac{k}{k_0} \right)^{n_s-1}$

Tensors : $\frac{k^3}{2\pi^2} P_T(k) = \frac{2}{\pi^2} \frac{H^2}{M_{Pl}^2} \equiv A_t \left(\frac{k}{k_0} \right)^{n_t}$



III. CORRELATORS



$$\left\langle \prod_{a=1}^N \phi(\underline{x}_a) \right\rangle \equiv \left\langle \Omega(\eta_*) \left| \prod_{a=1}^N \hat{\Phi}(\eta_*, \underline{x}_a) \right| \Omega(\eta_*) \right\rangle$$

OPERATOR STATE

IN-IN FORMALISM = Canonical Quantization

1) Evolution in the "interaction picture":

$$H \equiv H_0 + H_{\text{int}}$$

- Operators evolve with $H_0 \leftarrow$ free field mode fcts
- States evolve with H_{int} :

$$\begin{aligned} |\Omega(\eta_*)\rangle &= \hat{U}(\eta_*, -\infty) |\Omega(-\infty)\rangle \\ &= \text{TIME ORDERING} \int_{-\infty}^{\eta_*} d\eta \hat{H}_{\text{int}}(\eta) \text{ INITIAL STATE} |\Omega(-\infty)\rangle \end{aligned}$$

2) Projection onto the vacuum

$$\eta_i \equiv -\infty(1 - i\epsilon) \quad |\Omega(\eta_i)\rangle \rightarrow |0\rangle$$

INTERACTING FREE

PROOF:

$$\text{Let } |0\rangle \equiv \sum_N |N\rangle \langle N|0\rangle$$

(energy eigenstates

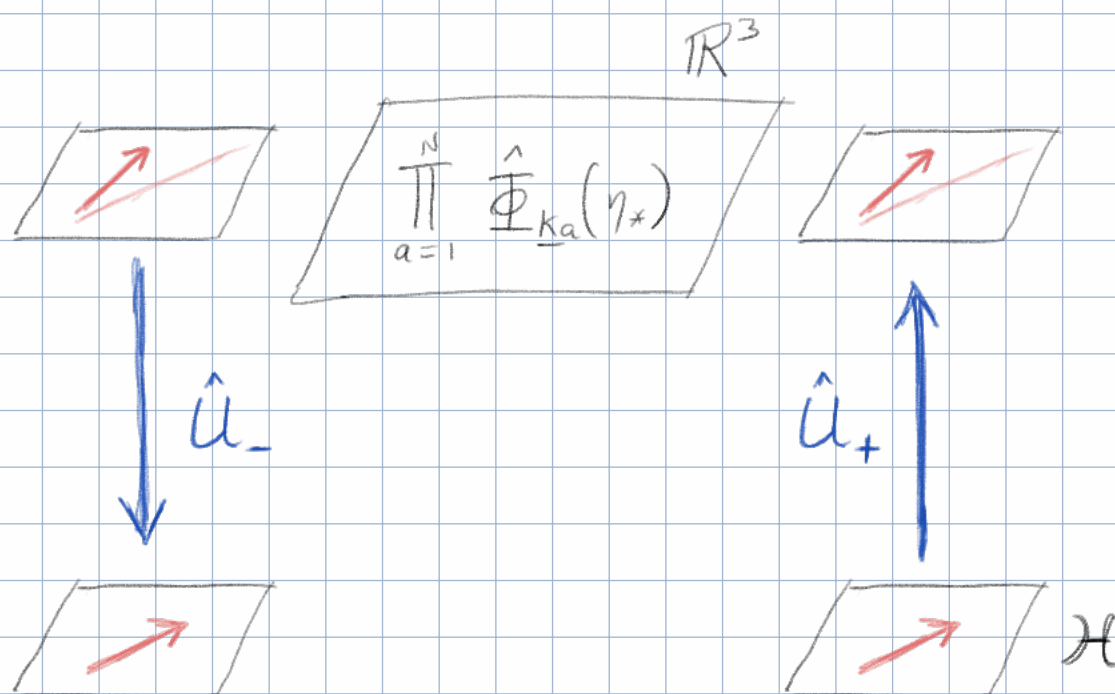
$$\begin{aligned}
 e^{-i\hat{H}(\eta-\eta_i)} |0\rangle &= \sum_N e^{-iE_N(\eta-\eta_i)} |N\rangle \langle N|0\rangle \\
 &= e^{-iE_\Omega(\eta-\eta_i)} |\Omega\rangle \langle \Omega|0\rangle + \sum_{\tilde{N} \neq \Omega} e^{-iE_{\tilde{N}}(\eta-\eta_i)} |\tilde{N}\rangle \langle \tilde{N}|0\rangle \\
 &\approx e^{-iE_\Omega(\eta-\eta_i)} |\Omega\rangle \langle \Omega|0\rangle
 \end{aligned}$$

$\eta_i \equiv -\infty(1-i\epsilon)$
 $e^{-\infty \times \epsilon E_{\tilde{N}}} \rightarrow 0$

L

3) Master formula

$$\left\langle \prod_{a=1}^N \phi_{\underline{k}_a} \right\rangle = \langle 0 | \bar{T} e^{i \int_{-\infty}^{\eta_*} d\eta \hat{H}_{\text{int}}} \prod_{a=1}^N \hat{\Phi}_{\underline{k}_a}(\eta_*) T e^{-i \int_{-\infty}^{\eta_*} d\eta \hat{H}_{\text{int}}} |0\rangle$$



Substitute mode expansion

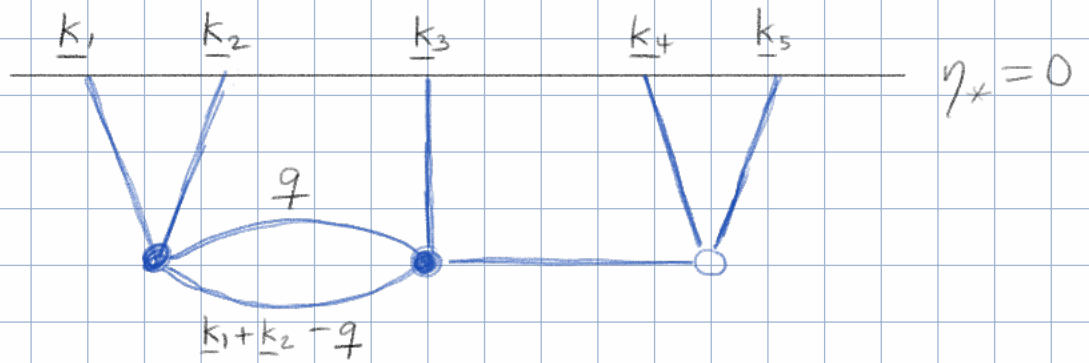
$$\hat{\Phi}(\eta, \underline{x}) = f_{\underline{k}}(\eta) \hat{a}_{\underline{k}} + f_{\underline{k}}^*(\eta) \hat{a}_{-\underline{k}}^+$$

and expand.

4) FEYNMAN RULES

To compute an N -point function:

- Draw a graph:



- Color each vertex black or white. Sum over all colorings.
- Assign a BULK-TO-BULK propagator to each INTERNAL LINE:

$$\bullet \text{---} \bullet = G_{++}(k; \eta_1, \eta_2) = W(k; \eta_1, \eta_2) \Theta(\eta_1 - \eta_2) + W^*(k; \eta_1, \eta_2) \Theta(\eta_2 - \eta_1)$$

$$\circ \text{---} \circ = G_{--}(k; \eta_1, \eta_2) = G_{++}^*(k; \eta_1, \eta_2)$$

$$\bullet \text{---} \circ = G_{+-}(k; \eta_1, \eta_2) = W^*(k; \eta_1, \eta_2) \quad \text{WIGHTMAN FUNCTION}$$

$$\circ \text{---} \bullet = G_{-+}(k; \eta_1, \eta_2) = W(k; \eta_1, \eta_2) \equiv f_k(\eta_1) f_k^*(\eta_2)$$

- Assign a BULK-TO-BOUNDARY propagator to each EXTERNAL LINE:

$$\text{---} \bullet = K_+(k; \eta, \eta_*) = W^*(k; \eta, \eta_*)$$

$$\text{---} \circ = K_-(k; \eta, \eta_*) = W(k; \eta, \eta_*)$$

- Assign a VERTEX FACTOR to each VERTEX:

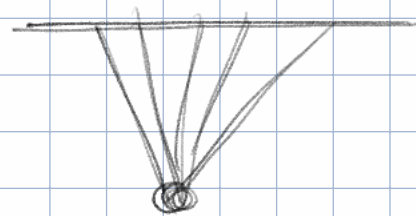
$$\text{---} \bullet \text{---} \text{---} = -i \int d\eta \sqrt{-g} V$$

$$\text{---} \circ \text{---} \text{---} = i \int d\eta \sqrt{-g} V$$

- Integrate over LOOP MOMENTA,

EXAMPLE

Consider $\dot{\Phi}^N$ in de Sitter



$$f_k(\eta) = \frac{iH}{\sqrt{2k^3}} (1 + ik\eta) e^{-ik\eta}$$

$$f'_k(\eta) = iH(k_\eta^2) \frac{1}{\sqrt{2k^3}} e^{-ik\eta}$$

$$\dot{f}_k = \frac{1}{a} f'_k = (-H\eta) f'_k$$



$$= f_k(\eta_*) f_k^*(\eta)$$



$$+ \text{ } \img alt="Diagram of a vertex with N lines radiating upwards to a horizontal line." data-bbox="290 391 410 462"/> = 2 \text{Re} \left[\img alt="Diagram of a vertex with N lines radiating upwards to a horizontal line." data-bbox="630 391 750 462"/> \right]$$

$$= 2 \text{Re} \left[-i\lambda \int_{-\infty}^0 \frac{d\eta}{(H\eta)^4} (-H\eta)^N \prod_{a=1}^N \partial_\eta (f_{k_a}(\eta_*) f_{k_a}^*(\eta)) \right]$$

$$\sim \lambda \text{Re} \left[-i H^{N-4} \int_{-\infty}^0 d\eta \eta^{N-4} \prod_{a=1}^N (-1)^N \frac{H^2}{k_a^3} k_a^2 \eta e^{ik_a \eta} \right]$$

$$= \left[\frac{\lambda H^{3N-4}}{\prod_{a=1}^N k_a} \frac{1}{E^{2N-3}} \right]$$

$$E \equiv \sum_{a=1}^N k_a$$

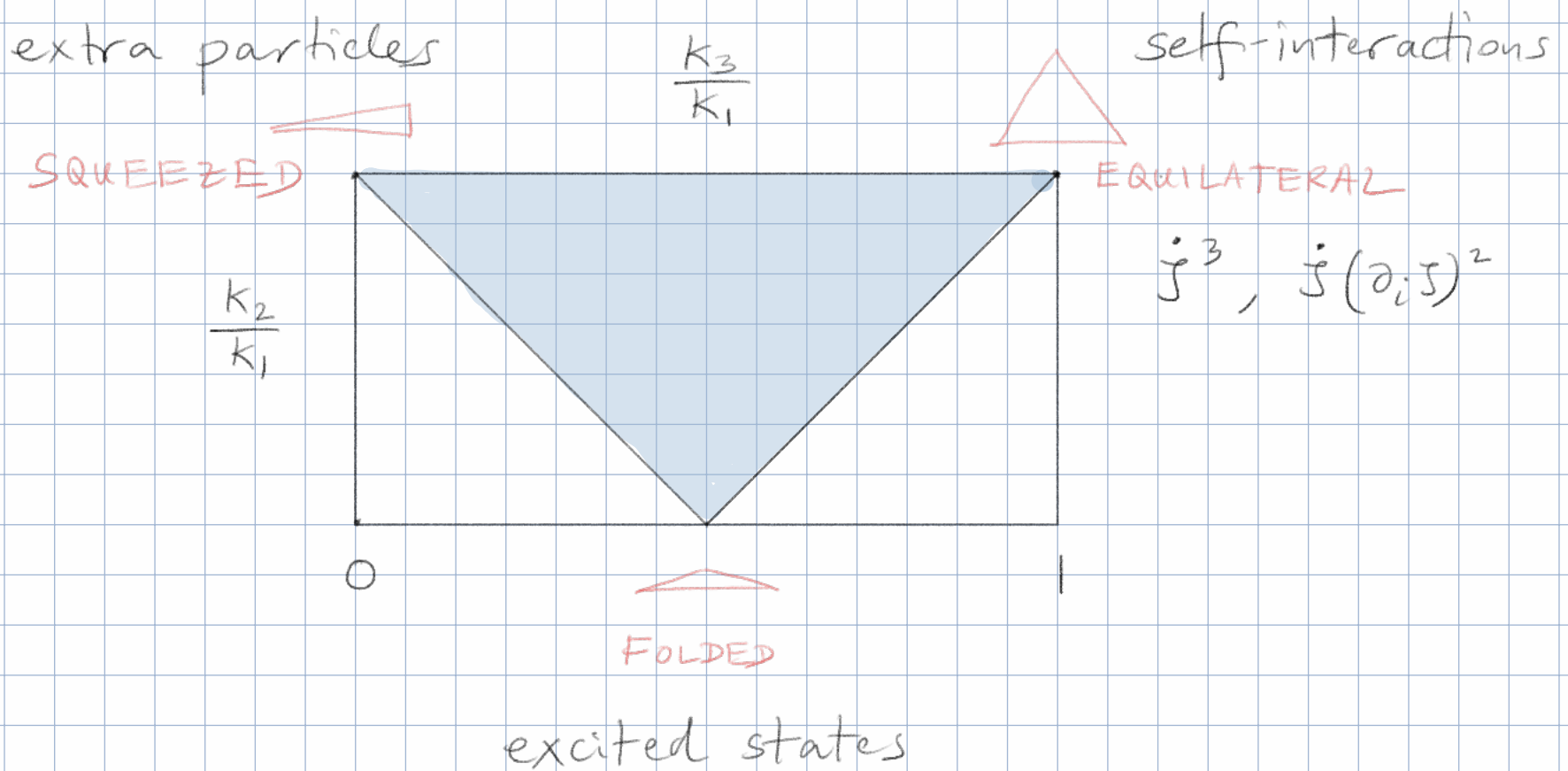
TOTAL ENERGY

using $\int_{-\infty(1-i\epsilon)}^0 d\eta e^{iE\eta} = \frac{1}{iE}$

$$\int_{-\infty(1-i\epsilon)}^0 d\eta \eta^p e^{iE\eta} = (-i\partial_E)^p \int_{-\infty(1-i\epsilon)}^0 d\eta e^{iE\eta}$$

INFLATIONARY BISPECTRA

$$\langle \zeta_{\underline{k}_1} \zeta_{\underline{k}_2} \zeta_{\underline{k}_3} \rangle = (2\pi)^3 \delta(\underline{k}_1 + \underline{k}_2 + \underline{k}_3) B_{\zeta}(k_1, k_2, k_3)$$



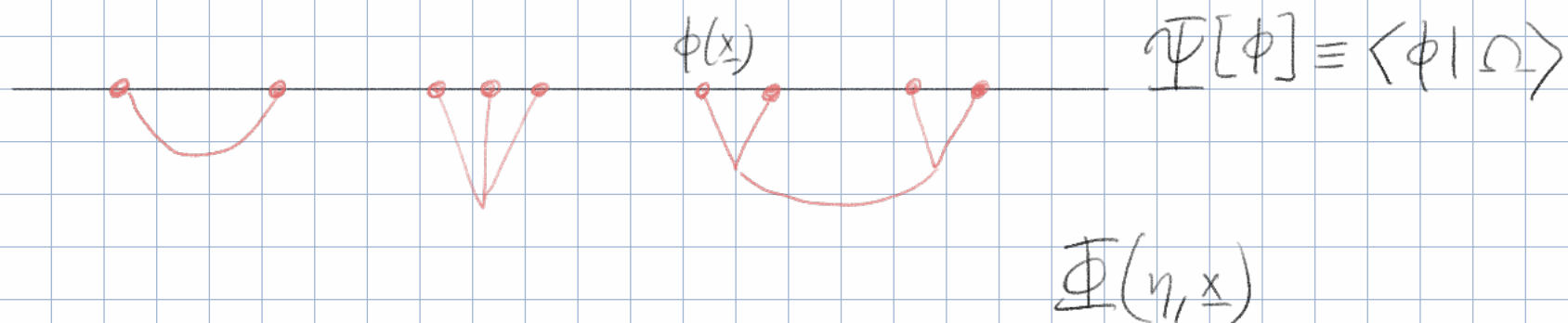
- amplitude:

$$f_{NL} = \frac{5}{18} \frac{B_{\zeta}(k, k, k)}{P_{\zeta}(k)^2} < 10-100$$

- slow-roll inflation: $f_{NL} = \frac{5}{12} \left(g\epsilon + \frac{\dot{\epsilon}}{H\epsilon} \right)$
- single-field consistency relation:

$$\lim_{k_3 \rightarrow 0} B_{\zeta}(k_1, k_2, k_3) = (1-n_s) P_{\zeta}(k_1) P_{\zeta}(k_3)$$

IV. WAVEFUNCTION



In the Schrödinger picture, the state evolves as

$$|\Omega(\eta)\rangle = \hat{U}(\eta, -\infty) |\Omega(-\infty)\rangle$$

Let $|\phi\rangle$ be an eigenstate of $\hat{\Phi}(\eta, \underline{x})$:

$$\hat{\Phi}(\eta_*, \underline{x}) |\phi\rangle = \phi(\underline{x}) |\phi\rangle$$

The wavefunction $\langle \phi |$ is

$$\Psi[\phi] \equiv \langle \phi | \Omega \rangle$$

$$= \langle \phi | T e^{-i \int_{-\infty}^{\eta_*} d\eta \hat{H}_{int}(\eta)} | \Omega(-\infty(1+i\epsilon)) \rangle = \int_{|0\rangle}^{| \phi \rangle} [d\Phi] e^{i S[\Phi]}$$

Correlators follow from the BORN RULE:

$$\left\langle \prod_{a=1}^N \phi_{\underline{k}_a} \right\rangle = \int [d\phi] \left(\prod_{a=1}^N \phi_{\underline{k}_a} \right) |\Psi[\phi]|^2$$

FREE THEORY

- Gaussian wavefunction $\Psi[\phi] = N e^{-\frac{1}{2} \int d^3k \gamma_2(k) \phi_k \phi_{-k}}$
- Two-point function:

$$\begin{aligned} \langle \phi_k \phi_{k'} \rangle &\sim \int [d\phi] \phi_k \phi_{k'} |e^{-\frac{1}{2} \int \gamma_2 \phi^2}|^2 \\ &= (2\pi)^3 \delta(k+k') \frac{1}{2 \operatorname{Re} \gamma_2(k)} \stackrel{=}{=} P_\phi(k) \end{aligned}$$

- Stationary phase:

$$\Psi[\phi] \approx N e^{iS[\Phi_{cl}]}$$

where $\Phi_{cl}(\eta, \underline{k})$ is a classical solution with $\begin{cases} \Phi_{cl}(\eta, \underline{k}) = \phi_k \\ \Phi_{cl}(\eta \rightarrow -\infty(1+i\epsilon), \underline{k}) = 0 \end{cases}$

$$\Phi_{cl}(\eta, \underline{k}) = \frac{f_k^*(\eta)}{f_k^*(\eta_*)} \phi_k$$

$$\log \Psi = iS[\Phi_{cl}]$$

$$= i \int d^3x d\eta a^4(\eta) \left(-\frac{1}{2} (\partial_\mu \Phi_{cl})^2 \right)$$

$$= i \int \frac{d^3k}{(2\pi)^3} d\eta a^2(\eta) \left(\frac{1}{2} \partial_\eta \Phi_{cl}(\underline{k}) \partial_\eta \Phi_{cl}(-\underline{k}) + \frac{1}{2} k^2 |\Phi_{cl}(\underline{k})|^2 \right)$$

$$= i \int \frac{d^3k}{(2\pi)^3} d\eta \cancel{(\text{EOM})} + \frac{i}{2} \int \frac{d^3k}{(2\pi)^3} d\eta \partial_\eta (a^2 \Phi_{cl} \partial_\eta \Phi_{cl})$$

$$= \frac{i}{2} \int \frac{d^3k}{(2\pi)^3} (a^2 \Phi_{cl} \Phi'_{cl})_{\eta=\eta_*}$$

$$\equiv -\frac{1}{2} \int \frac{d^3k}{(2\pi)^3} \gamma_2(\underline{k}) \phi_k \phi_{-k}$$

$$\boxed{\gamma_2(\underline{k}) = \frac{-i}{(H\eta_*)^2} \left. \frac{\partial_\eta f_k^*}{f_k^*} \right|_{\eta=\eta_*}}$$

Recall that

$$f_k^*(\eta) = \frac{-iH}{\sqrt{2k^3}} (1 - ik\eta) e^{ik\eta} \quad \partial_\eta f_k^*(\eta) = \frac{-iH}{\sqrt{2k^3}} k^2 \eta e^{ik\eta}$$

so that

$$\begin{aligned} \psi_2(\underline{k}) &= \frac{-i}{(H\eta_*)^2} \frac{k^2 \eta_*}{1 - ik\eta_*} \cdot \frac{1 + ik\eta_*}{1 + ik\eta_*} \\ &= \frac{1}{1 + (k\eta_*)^2} \left[\frac{k^3}{H^2} - i \frac{k^2}{H^2 \eta_*} \right] \end{aligned}$$

divergent, but imaginary

Hence, we get

$$\boxed{P_\phi(k) = \frac{H^2}{2k^3}} \quad \checkmark$$

INTERACTIONS

$$\Psi[\phi] = e^{-\frac{1}{2} \int \psi_2 \phi^2} \exp \left[\sum_{n=3}^{\infty} \int \dots \int \gamma_n \phi_{\underline{k}_1} \dots \phi_{\underline{k}_n} \delta(\underline{k}_1 + \dots + \underline{k}_n) \right]$$

GAUSSIAN

WAVEFUNCTION COEFFICIENTS

$$\begin{aligned} \left\langle \prod_{a=1}^N \phi_{\underline{k}_a} \right\rangle &= \int [d\phi] |\Psi[\phi]|^2 \left(\prod_{a=1}^N \phi_{\underline{k}_a} \right) \\ &= \int [d\phi] \left| e^{-\frac{1}{2} \int \psi_2 \phi^2} \right|^2 \left| 1 + \int \gamma_3 \phi^3 + \int \gamma_4 \phi^4 + \dots \right. \\ &\quad \left. \dots \frac{1}{2} \left(\int \gamma_3 \phi^3 \right)^2 + \dots \right|^2 \left(\prod_{a=1}^N \phi_{\underline{k}_a} \right) \end{aligned}$$

Do lots of Gaussian integrals!

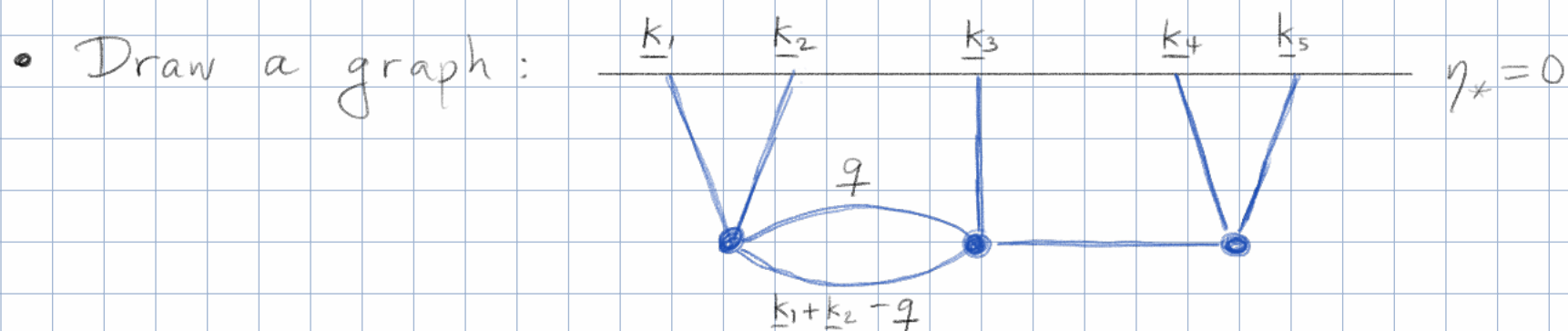
For example:

$$\left\langle \prod_{a=1}^3 \phi_{\underline{k}_a} \right\rangle = \frac{\psi_3 + \psi_3^*}{\prod_{a=1}^3 2 \operatorname{Re} \psi_2} + O(\psi_5, \psi_3^3, \dots)$$

$$\left\langle \prod_{a=1}^4 \phi_{\underline{k}_a} \right\rangle = \frac{\psi_4 + \psi_4^*}{\prod_{a=1}^4 2 \operatorname{Re} \psi_2} + \frac{\psi_3^* \psi_3}{\left(\prod_{a=1}^4 2 \operatorname{Re} \psi_2 \right) \operatorname{Re} \psi_2(k_I)}$$

FEYNMAN RULES

To compute an N -point wavefunction coefficient:



- Assign a BULK-TO-BOUNDARY propagator to each EXTERNAL LINE:

$$\text{---} \bullet \equiv K(k, \eta) = \frac{f_k^*(\eta)}{f_k^*(\eta_*)}$$

- Assign a BULK-TO-BULK propagator to each INTERNAL LINE:

$$\bullet \text{---} \bullet \equiv G(k; \eta_1, \eta_2)$$

$$= \underbrace{f_k(\eta_1) f_k^*(\eta_2) \Theta(\eta_1 - \eta_2) + f_k(\eta_2) f_k^*(\eta_1) \Theta(\eta_2 - \eta_1)}_{\text{FEYNMAN}} - \frac{f_k(\eta_*)}{f_k^*(\eta_*)} f_k^*(\eta_1) f_k^*(\eta_2)$$

- Assign a VERTEX FACTOR $\text{---} \bullet \text{---} \text{---} = i \int d\eta \sqrt{-g} V$

EXERCISE

Derive γ_3 for $\dot{\Phi}^3$ in dS_4 .

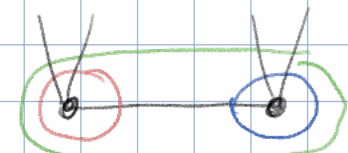
Show that it gives the correct $\langle \phi_{\underline{k}_1} \phi_{\underline{k}_2} \phi_{\underline{k}_3} \rangle$.

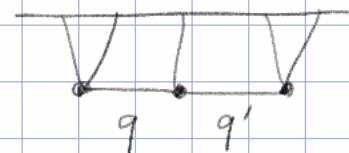
EXAMPLE: Flat-Space Wavefunction

$f_k(\eta) = \frac{1}{\sqrt{2k}} e^{-ik\eta}$ leads to simple integrals

Consider $\lambda \dot{\Phi}^3$ in $\mathbb{R}^{1,3}$.

$$\gamma_3 = \text{V} = i\lambda \int_{-\infty}^{t_*} dt e^{i(\underbrace{k_1+k_2+k_3}_{\equiv K})(t-t_*)} = \frac{\lambda}{K}$$

$$\begin{aligned} \gamma_4 = \text{V} \text{---} \text{V} &= -\lambda^2 \int_{-\infty}^{t_*} dt \int_{-\infty}^{t_*} dt' e^{ik_{12}t} G(q, t, t') e^{ik_{34}t} \\ &= \frac{\lambda^2}{2K_I} \left[\frac{1}{EE_R} + \frac{1}{EE_L} - \frac{1}{E_LE_R} \right] \\ &= \frac{\lambda^2}{EE_LE_R} \end{aligned}$$


$$\gamma_5 = \text{V} \text{---} \text{V} \text{---} \text{V} = \frac{\lambda^3}{EE_LE_ME_R} \left[\frac{1}{k_{123} + q'} + \frac{1}{k_{345} + q} \right]$$


V. BOOTSTRAP

Derive wavefunction coefficients $\mathcal{T}_N \equiv \langle 0, 0_2 \dots 0_N \rangle$

using SYMMETRIES, UNITARITY, CAUSALITY and LOCALITY.

SYMMETRIES

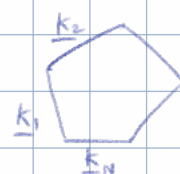
- $SO(4,1)$ {
- translations
 - rotations
 - dilatations
 - (• conformal)

$$\mathcal{T}_N(\underline{k}_a) = (2\pi)^3 \delta(\underline{k}_1 + \dots + \underline{k}_N) \psi_N(\underline{k}_a)$$

$$\psi_N(\lambda \underline{k}_a) = \lambda^{\sum \Delta_a - 3(N-1)} \psi_N(\underline{k}_a)$$

$$\Delta_a \equiv \frac{3}{2} + \sqrt{\frac{9}{4} - \frac{m_a^2}{H^2}}$$

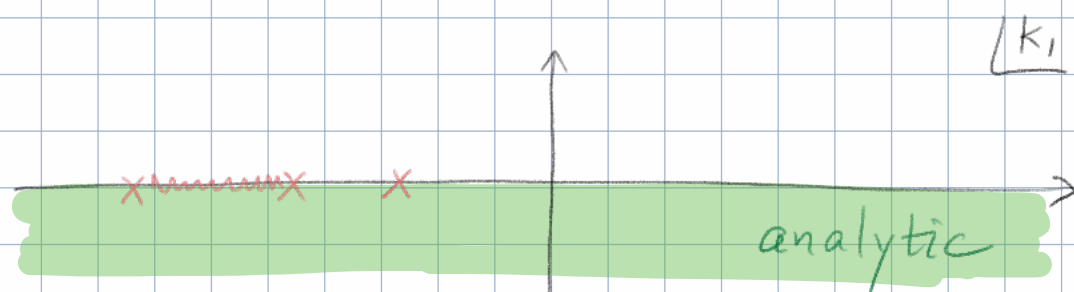
Ward identity: $\sum_{a=1}^N \left(k_a^\mu \partial_{\underline{k}_a}^\mu + \dots \right) \psi_N(\underline{k}_a) = 0$



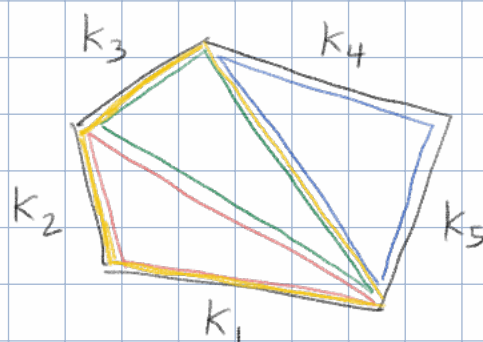
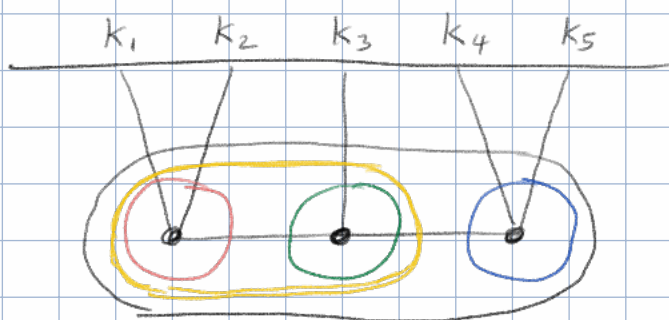
$k_a \equiv |\underline{k}_a|$ "energies"

CAUSALITY

is reflected in the ANALYTIC STRUCTURE of $\mathcal{T}_N$:



LOCALITY constrains the allowed SINGULARITIES:



UNITARITY constrains FACTORIZATION on singularities.

Consider the Schrödinger equation

$$i \partial_t \Psi[\phi, t] = \hat{H} \Psi[\phi, t]$$

$$\text{where } \hat{H} = \frac{1}{2} \iint_{\underline{k}_1, \underline{k}_2} \left(-\frac{\delta}{\delta \phi_{\underline{k}_1}} \frac{\delta}{\delta \phi_{\underline{k}_2}} + k_1^2 \phi_{\underline{k}_1} \phi_{\underline{k}_2} \right) + \lambda \iiint_{\underline{k}_1, \underline{k}_2, \underline{k}_3} \phi_{\underline{k}_1} \phi_{\underline{k}_2} \phi_{\underline{k}_3}.$$

Substitute the perturbative expansion of Ψ

$\Rightarrow$ infinite number of constraints on ψ_n 's:

$$\boxed{\tilde{\psi}_n(k_a) \equiv \psi_n(k_a) + \psi_n^*(-k_a) \equiv - \sum_{\text{cuts}} P(q) \tilde{\psi}_m \tilde{\psi}_{n-m}}$$

For rational correlators, this fixes much of the answer.

EXAMPLE : Φ^3 in $\mathbb{R}^{1,3}$

$$\text{Diagram: a triangle with a horizontal line at the top and a vertical line at the bottom, with a red dot on the bottom line.} = -P(q) \text{Diagram: a triangle with a horizontal line at the top and a vertical line at the bottom, with a blue dot on the bottom line.} \times \text{Diagram: a triangle with a horizontal line at the top and a vertical line at the bottom, with a blue dot on the bottom line.}$$

$$\tilde{\psi}_4(k_a, q) = -P(q) \tilde{\psi}_3(k_{12}, q) \tilde{\psi}_3(k_{34}, q)$$

$$\begin{aligned} \text{where } \tilde{\psi}_3(k_{12}, q) &= \psi_3(k_{12}, q) + \psi_3^*(-k_{12}, q) \\ &= \frac{1}{k_{12} + q} + \frac{1}{-k_{12} + q} = -\frac{2q}{k_{12}^2 - q^2} \end{aligned}$$

Hence, we have

$$\tilde{\psi}_4(k_a, q) = -\frac{2q}{(k_{12}^2 - q^2)(k_{34}^2 - q^2)}$$

Consider the limits

$$\begin{aligned}\lim_{E_L \rightarrow 0} \tilde{\psi}_4(k_a, q) &= \lim_{E_L \rightarrow 0} (\psi_4(k_a, q) + \psi_4^*(-k_a, q)) \\ &= \lim_{E_L \rightarrow 0} \psi_4(k_a, q) = \frac{\cancel{-2q}}{(k_{12} + q) \cancel{(-2q)} (k_{34}^2 - q^2)} \\ &= \frac{1}{E_L} \frac{1}{(k_{34} + q)(k_{34} + k_{12})}\end{aligned}$$

$$\lim_{E_R \rightarrow 0} \psi_4(k_a, q) = \frac{1}{E_R} \frac{1}{(k_{12} + q)(k_{12} + k_{34})}$$

Unique solution is

$$\boxed{\psi_4 = \frac{1}{E_L E_R E}} \quad \checkmark$$

Note that

$$\begin{aligned}\lim_{E \rightarrow 0} \psi_4 &= \frac{1}{E} \frac{1}{(k_{12} + q)(-k_{12} + q)} \\ &= -\frac{1}{E} \frac{1}{k_{12}^2 - q^2} \equiv -\frac{A_4}{E}\end{aligned}$$

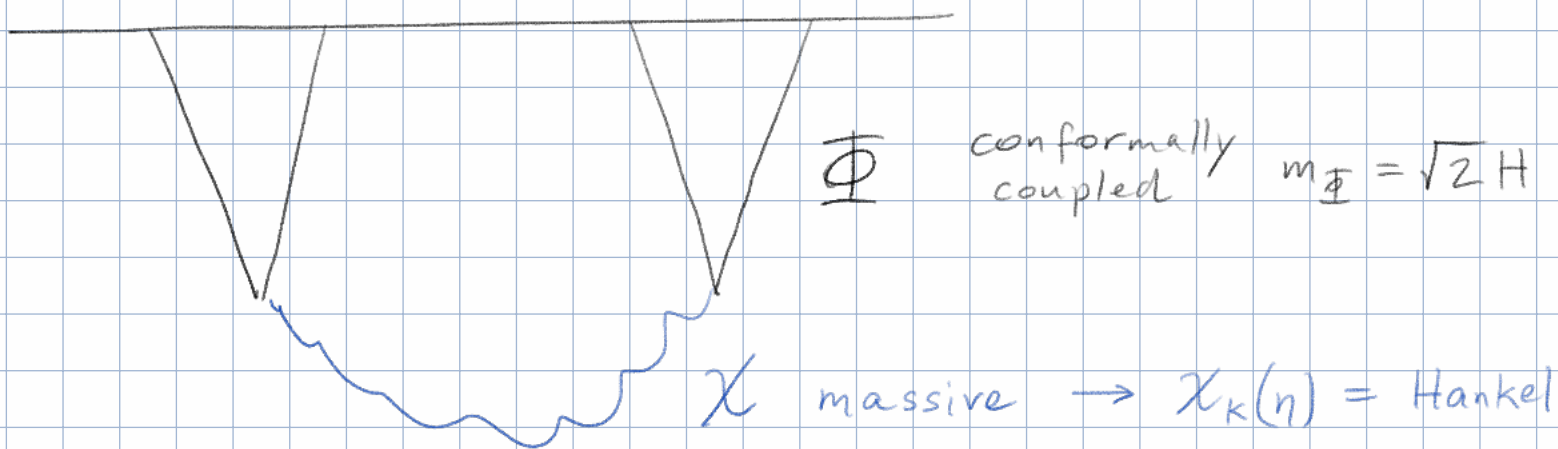
MANDELSTAM

EXERCISE: $\dot{\Phi}^3$ in dS_4

Show that

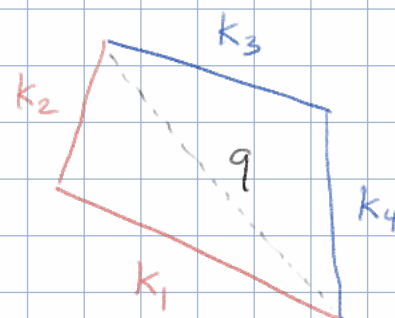
$$\begin{aligned}\psi_4 &= 2q(k_1 k_2 k_3 k_4)^2 \left[\frac{6}{E^5} \left(\frac{1}{E_L} + \frac{1}{E_R} \right) + \frac{3}{E^4} \left(\frac{1}{E_L^2} + \frac{1}{E_R^2} \right) \right. \\ &\quad \left. + \frac{1}{E^3} \left(\frac{1}{E_L^3} + \frac{1}{E_R^3} \right) - \frac{1}{E_L^3 E_R^3} \right]\end{aligned}$$

COSMOLOGICAL COLLIDER



Direct computation is hard $\rightarrow$ Bootstrap

Let $u \equiv \frac{q}{k_1 + k_2}$, $v \equiv \frac{q}{k_3 + k_4}$



$$\psi_4(k_a, q) = \frac{1}{q} \hat{\psi}_4(u, v)$$

Ward identity:

$$\left[u^2(1-u^2) \partial_u^2 - 2u^3 \partial_u + \frac{M^2}{H^2} \right] \hat{\psi}_4 = \frac{uv}{u+v}$$

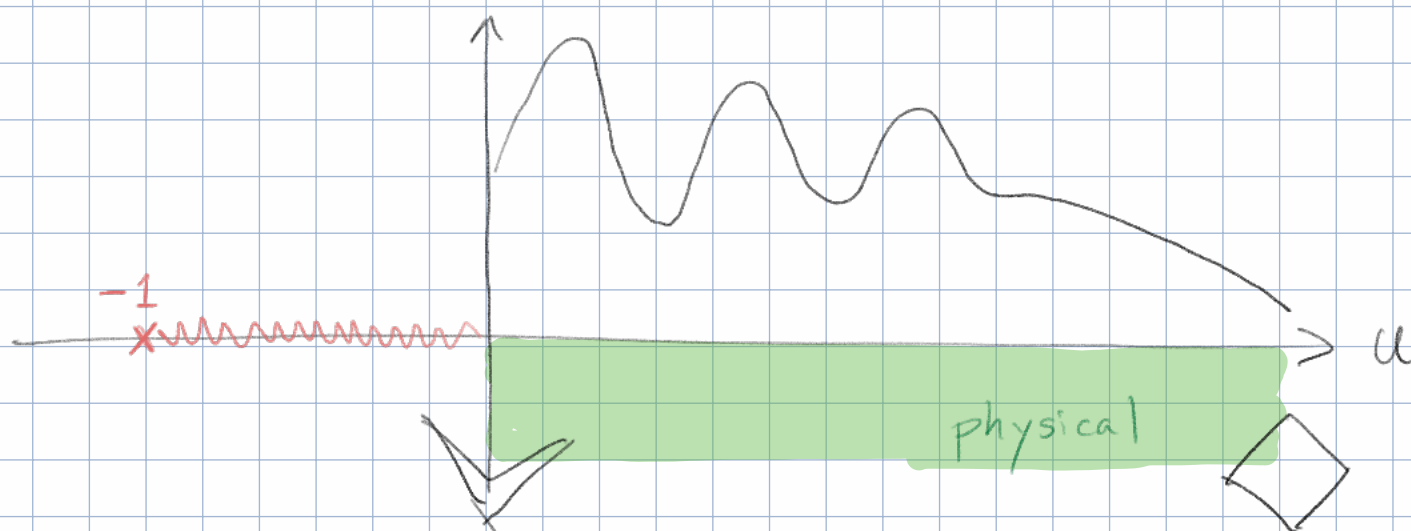
Hypergeometric Equation

Collapsed limit: $u \rightarrow 0$

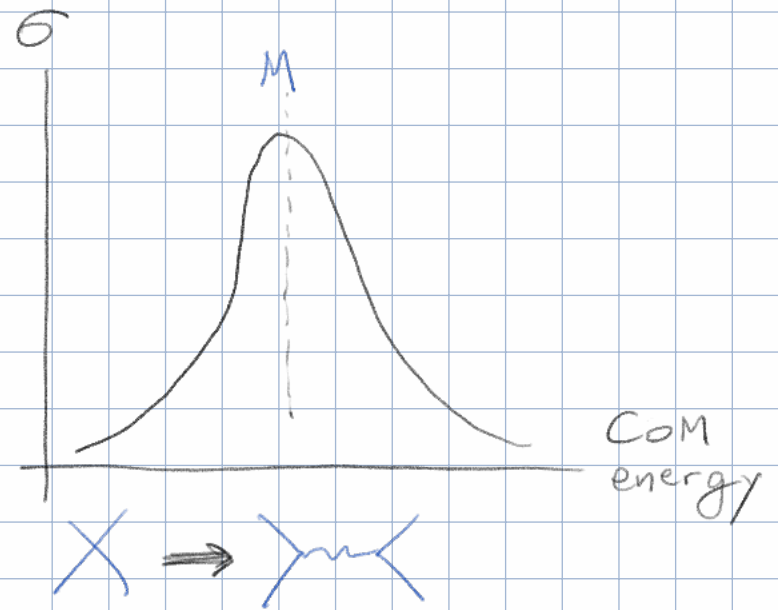
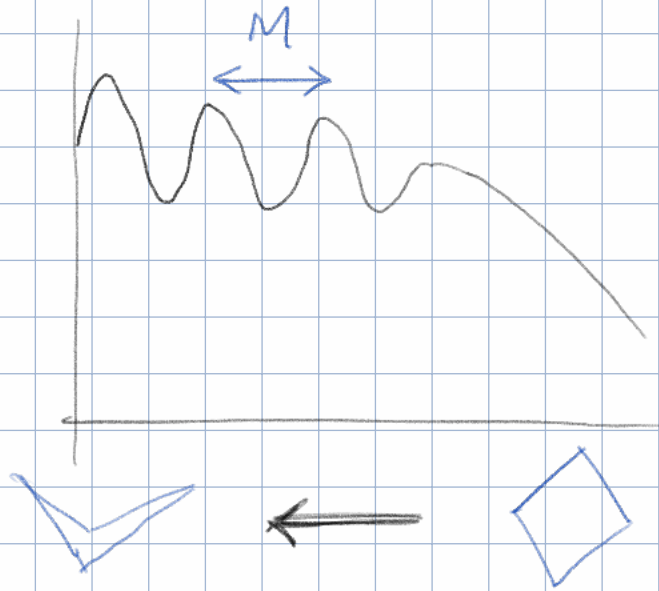


$$\lim_{u \rightarrow 0} \hat{\psi}_4 \sim u^{iM/H} = e^{iM/H \log u}$$

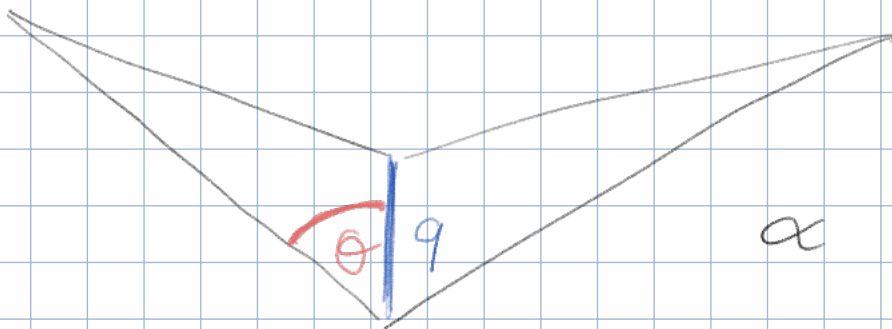
OSCILLATIONS



Oscillations are the analog of a RESONANCE in collider physics:



Perform PARTICLE SPECTROSCOPY up to $E \sim 10^{14}$ GeV :



$$\propto \sin(\underbrace{M}_{\text{MASS}} \log q) \underbrace{P_J}_{\text{SPIN}}(\cos \theta)$$

Inflation was the ultimate collider :

- Correlations record the physics of inflation.
- New particles leave distinctive signals.
- Upcoming galaxy surveys will search for these signals.

THE END